Psychedelic Bliss
A new psychedelic drug could treat depression and addiction, without the hallucinations. **http://dgit.in/jan21-15**

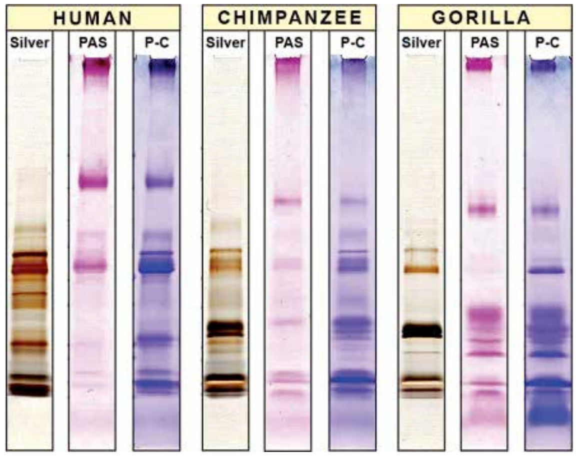
Body-snatching Fungi
A new species of deadly fungi turns flies into zombies and devours them from inside out. **http://dgit.in/jan21-16**

amount of amylase that they do. Stefan Ruhl, one of the researchers on the team has a theory though. The idea is that the partial breakdown of starchy food allowed humans to taste sugar, which they otherwise could not have, which in turn pushed them towards finding more starch rich foodstuff. Ruhl explains, "unlike the baboons who predigest food in their cheek pouches, we humans do not keep food in our mouths long enough for any substantial digestion to happen. One idea is that salivary amylase evolved to help our ancestors detect starch: They would not be able to taste it otherwise. Amylase liberates sugar in starch, and this may help animals develop a taste preference for starch-rich foods like potatoes or corn."

DIFFERENCES FROM THE APES

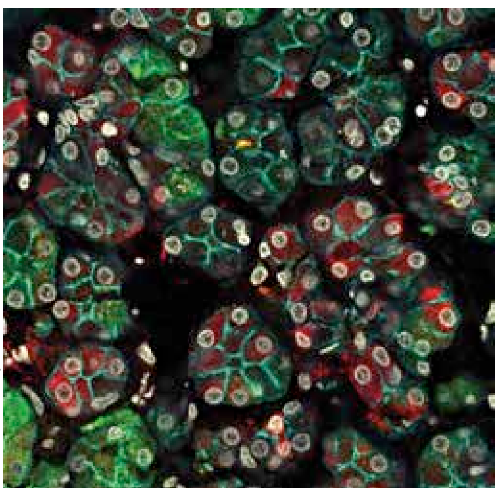
The team looked more closely at the differences between the saliva of human beings and that of their closest relatives, chimpanzees and gorillas. Instead of looking at just a single compound, the researchers compared the presence of all proteins present in the saliva samples. There are thousands of these proteins. Despite the close genetic proximity to chimpanzees and gorillas, turns out that humans have a very distinctive mix of salivary proteins. The finding seems to indicate that over two million years of specialised diet, where humans have diverged considerably from their closest animal relatives, may have caused the changes in the salivary proteins.

The study also showed that humans have a far more watery saliva than chimpanzees, gorillas or macaques. The protein content in human saliva was considerably reduced compared to their wild cousins. Humans also had a larger amount of amylase, as well as anhydrase VI. Both of these proteins are believed to play a role in taste perception. Humans and chimpanzee saliva had more of a protein that helped with fat metabolism, than gorillas. Together, the proteins in human saliva are tuned towards breaking down starches and fats. The saliva of the great apes had a lot of a protein known as latherin, which makes the saliva foamy. Humans did not have large concentrations of this protein, which the researchers believe is related to the loss of body hair in humans. Humans, unlike the great apes, do not engage in social grooming, and as such do not require latherin. Investigating the evolutionary processes that gave rise to the genetic machinery needed to produce each of these thousands of proteins, remains an area of active research, one that could provide surprising insights into our species.



Credit: Stefan Ruhl

Differences in Human, Chimpanzee and Gorilla salivary proteins, gathered using three different straining methods



Credit: Alison May

Microscoping imaging of submandibular cells has shown differences in cells that were thought to be very similar

HOW IS SALIVA PRODUCED

There are three pairs of glands in the mouth that produce saliva, which are the parotid, submandibular and sublingual glands. In their most recent study, the team of researchers investigated which of the proteins were secreted by which of the glands. As a consequence of the study, the researchers also identified the proteins in human saliva that do not originate from any of the cells in the glands, but instead have other origins, such as the epithelial membranes or blood plasma. The research also helped the scientists probe the exact differences between the three types of salivary glands, as well as their functions. For example, one of the findings of the study is that the parotid and submandibular glands produce a lot of amylase, but the sublingual glands make almost none. The research also indicated a new level of cellular diversity in the makeup of the glands, because apparently similar cells were producing distinctly different proteins. The research provides valuable insight into how the saliva production mechanism is supposed to work during the regular functioning of the body, and a snapshot of which proteins are expected to be present under normal conditions. This study provides a reference snapshots for future diagnostic approaches, that can be used to find signs of diseases in saliva. The study also indicates the immense complexity of saliva, and that it still holds many secrets waiting to be unlocked.